

Accountable Audit Summary

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Version 2.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Accountable on Accountable. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated October 16, 2025, was delivered to Accountable, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 13 days, the Cyfrin team conducted an audit on the Accountable code provided by Accountable. The findings consist of 4 Critical, 2 High, 12 Medium & 5 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Accountable
Repository	credit-vaults-internal
Commit	fc43546fe671...
Fix Commit	1ae7e2fb74a3...
Audit Timeline	Sep 22nd - Oct 8th, 2025
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	4	4	0
High Risk	2	2	0
Medium Risk	12	11	1
Low Risk	5	5	0
Informational	9	8	1
Gas Optimizations	1	1	0
Total	33	31	2

6 Audit Scope

```
src/access/AccessBase.sol
src/access/Authorizable.sol
src/access/Whitelistable.sol
src/constants/Errors.sol
src/factory/AsyncVaultFactory.sol
src/factory/FixedTermFactory.sol
src/factory/OpenTermFactory.sol
src/factory/RewardsFactory.sol
src/factory/StrategyFactoryBase.sol
src/modules/FeeManager.sol
src/modules/GlobalRegistry.sol
src/rewards/Rewards.sol
src/rewards/RewardsDistributorMerkle.sol
src/rewards/RewardsDistributorStrategy.sol
src/strategies/AccountableFixedTerm.sol
src/strategies/AccountableOpenTerm.sol
src/strategies/AccountableStrategy.sol
src/vault/AccountableAsyncRedeemVault.sol
src/vault/AccountableVault.sol
src/vault/queue/AccountableWithdrawalQueue.sol
```

A later commit 1d07a28 was also reviewed.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Accountable Loans provide fixed-term and open-term credit against pooled lender liquidity held in an "Accountable Async Vault". Lenders deposit the base asset to mint shares; a designated borrower draws and repays over time. In FixedTerm, interest is pre-scheduled per interval; in OpenTerm, a `_scaleFactor` (virtual share price) linearly accrues interest. Repayments (and, for FixedTerm, the periodic `pay`) route protocol, establishment and performance fees to a `FeeManager`, return net interest to the vault, and update accounting. Withdrawals are served instantly when liquid, or queued and later processed via a withdrawal queue. In FixedTerm, deposits occur during a defined deposit window; in OpenTerm, deposits are allowed while the loan is "OngoingDynamic". Vault shares entitle lenders to principal and accrued interest pro rata.

Key components:

- **AccountableAsync Vault and Withdrawal Queue:** lenders mint shares at a strategy-provided price (FixedTerm: usually 1e18 during the deposit period; OpenTerm: `_scaleFactor` outside the deposit period). `requestRedeem(shares)` queues withdrawals when liquidity is insufficient; depending on the vault mode, the queue pays at the share price when processed (CurrentPrice) or when requested (RequestPrice). Queued requests are processed as liquidity becomes available, with support for batched expiries and rollovers.
- **AccountableFixedTerm (strategy):** fixed terms (rate, intervals, duration, deposit window) that the borrower accepts; funds can be drawn up to capacity and minimum-cap constraints. Interest accrues per interval; `pay()` settles interest and fees, `repay()` reduces principal and `prepay()` settles early according to the terms.

Net interest is locked into the vault and fees go to the `FeeManager`. The strategy tracks `outstandingPrincipal`, `outstandingInterest`, `claimableInterest` and `drawableFunds`; lenders can claim interest (burn-for-yield model) or receive it via withdrawals.

- **AccountableOpenTerm (strategy):** the borrower accepts the terms to enter `OngoingDynamic`, during which deposits are allowed and the borrower can `borrow` and `supply` and later `repay`. `_scaleFactor` increases linearly with time and delinquency penalties; the share price equals `_scaleFactor` except during the initial deposit period or after repayment or default claims, when it references the vault's `assetShareRatio`. The loan tracks required liquidity against reserves to mark delinquency (with a grace period), and penalties accrue while delinquent. On repayment, assets are locked to the vault, queued withdrawals are processed, principal is reduced, configured fees are collected, and the loan moves to `Repaid` when principal reaches zero.
- **FeeManager:** calculates the performance, establishment and protocol fee split; strategies invoke `collect` during payments to route fees.

9 Attestation

Cyfrin attests that it performed a security review of the Accountable code provided by Accountable.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-10-16-cyfrin-accountable-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Accountable has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=5c7464d74ba4b457678d6366a3d94f4657f4a6e80be941fa5db60456180f4048
F=2025-10-16-cyfrin-accountable-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.